Project Proposal:Apex Pay

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1. Executive Summary

Apex Pay is a next-generation, hybrid financial ecosystem designed natively for Android. It bridges the gap between traditional banking and modern retail investing by merging three distinct platforms into a single, cohesive application: a Peer-to-Peer (P2P) digital wallet, a simulated retail brokerage, and an advanced Artificial Intelligence financial advisory layer.

2. Core Architecture & Tech Stack

The application relies on a dual-state architecture, balancing secure local data management with asynchronous network calls.

- **Frontend:** Android Studio (Java/XML) with custom rendering for translucent Glassmorphic UI elements.
- **Local Database:** SQLite for secure, offline-first management of the user's ledger, portfolio holdings, and transaction history.
- **Backend Integration: * Firebase Authentication:** For secure email and password onboarding.
    - **Firebase ML (Machine Learning):** For on-device computer vision and object recognition.
    - **Financial APIs:** To fetch real-time market tickers, crypto prices, and candlestick chart data.
    - **LLM API:** To power the conversational AI risk analyst.

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3. Detailed Feature Breakdown

3.1. Multi-Tiered Authentication & Security

The system ensures user data is protected through a two-step gatekeeping process.

- **Onboarding:** Users create an account utilizing Firebase Authentication (Email and Password). During setup, they establish a unique **Username** and are assigned a secure **Account Number**, alongside uploading a profile picture.
- **Daily Access:** To reduce friction, returning users bypass the login screen using the AndroidX BiometricPrompt API (FaceID/Fingerprint), with a 4-digit numeric PIN pad as a

secure fallback.

3.2. The Digital Wallet (P2P Transactions)

A fully functional local ledger system simulating a modern digital wallet like Venmo or CashApp.

- **Liquid Cash Management:** Users hold a primary balance stored securely in SQLite.
- **P2P Transfer Matrix:** A quick-send and add interface allowing users to search the network by **Username** or **Account Number**. It includes a vertical scrolling row of frequent contacts for instant money transfers.
- **Master Ledger:** A historical log of all platform activity (deposits, trades, P2P transfers) filterable by date and transaction type.

3.3. Virtual Card & Subscription Tracker

A module dedicated to expense management and automated billing tracking.

- **Interactive 3D Card:** A visually stunning, frosted-glass virtual debit card tied to the user's Account Number. Users can toggle a "Freeze/Unfreeze" state and reveal their CVV.
- **Recurring Expense Dashboard:** A dedicated list highlighting active monthly subscriptions (e.g., Netflix, Spotify), showing exact monthly costs and upcoming billing dates to prevent overdrafts.

3.4. Simulated Retail Brokerage

A dynamic trading environment powered by live market data.

- **Live Market Feeds:** Users can browse categorized lists (Crypto, Stocks, Top Gainers) displaying real-time prices and percentage changes.
- **Asset Deep-Dive:** Tapping an asset reveals an interactive Candlestick chart for price history, a Bar chart for daily trading volume, and core metrics like Market Cap and P/E Ratio.
- **Mock Execution Engine:** Users can simulate buying fractional shares or whole stocks using their "Liquid Cash." The system calculates the transaction, applies mock network fees, deducts the cash, and securely writes the new asset to the SQLite Portfolio table.

3.5. AI Financial Intelligence

The most advanced layer of the application, utilizing AI to educate and guide the user.

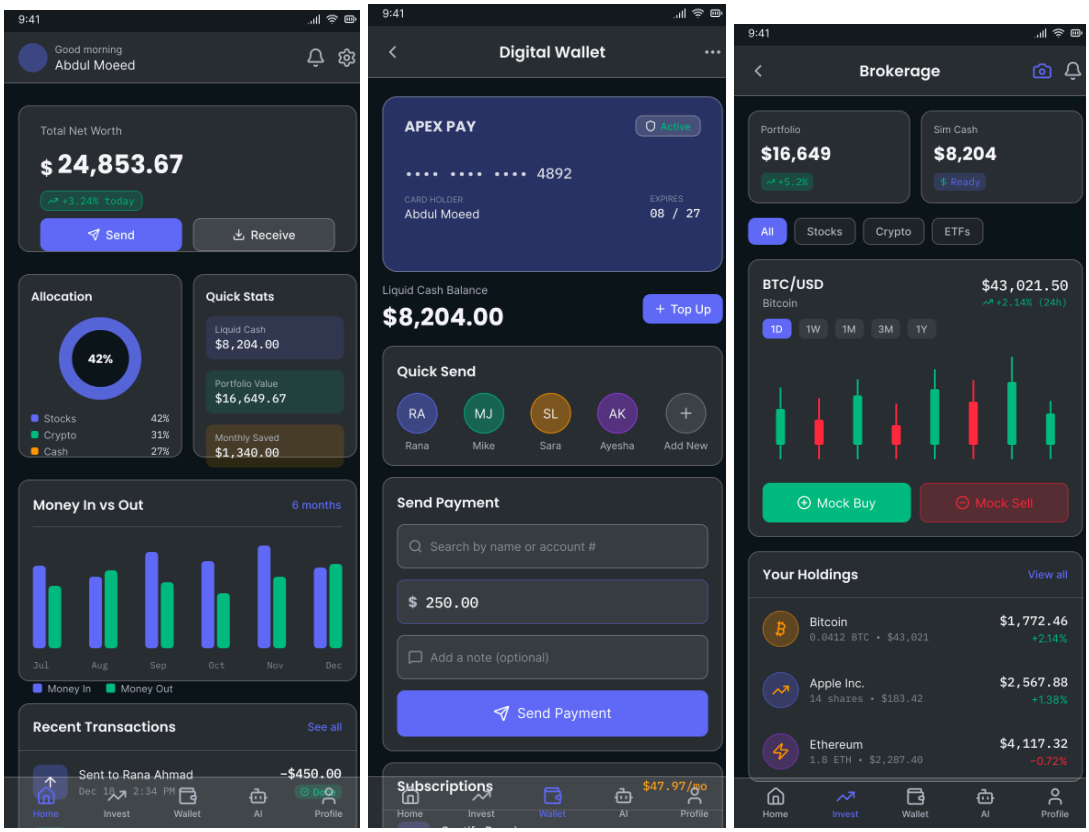
- **Scan-to-Invest (Computer Vision):** Utilizing Firebase ML, users can open their camera and point it at a physical product (e.g., a pair of Nike shoes or an Apple MacBook). The system recognizes the brand logo, maps it to its publicly traded ticker, and presents a one-tap bottom sheet to instantly buy shares.
- **Risk Analyst Chatbot:** A dedicated LLM-powered chat interface. Users can attach specific asset data to the chat and ask complex financial questions. The AI utilizes mathematical models (like Net Present Value and Discounted Cash Flow) to provide detailed, objective risk assessments.

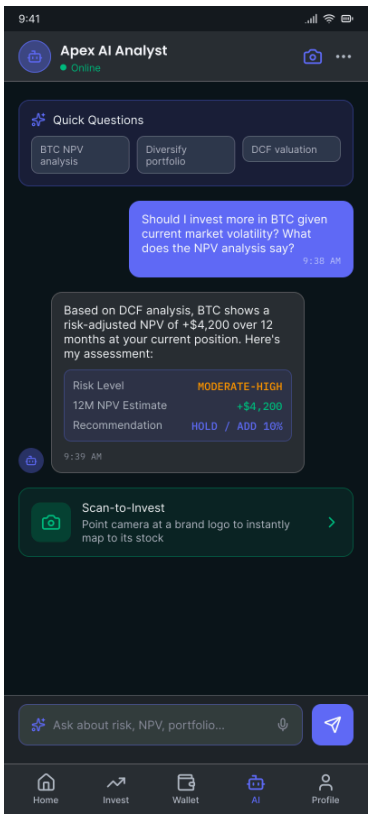
3.6. Visual Analytics Dashboard

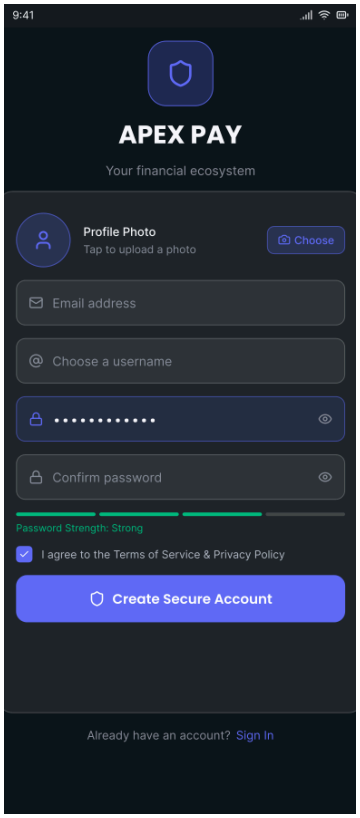
A data-rich environment for users to monitor their financial health.

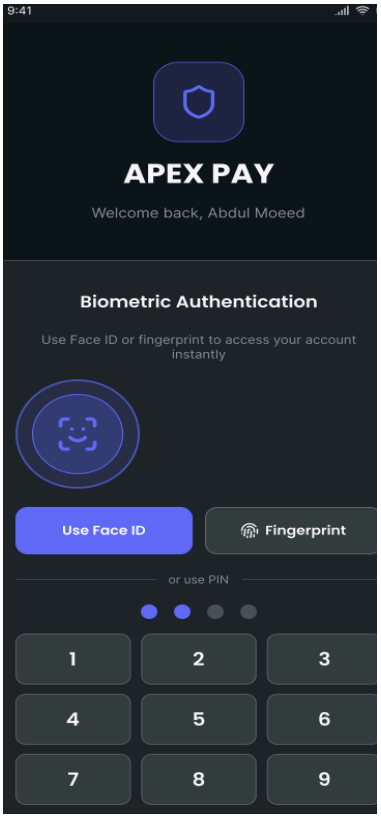
- **Portfolio Distribution:** A large, multi-colored Donut Chart visualizes the user's asset allocation (e.g., 60% Tech, 40% Crypto), with the total Profit/Loss anchored in the center.
- **Cash Flow Analysis:** A grouped Bar Chart tracks monthly spending habits, placing "Money In" (green bars) directly next to "Money Out" (red bars) for immediate visual feedback on monthly budgets.

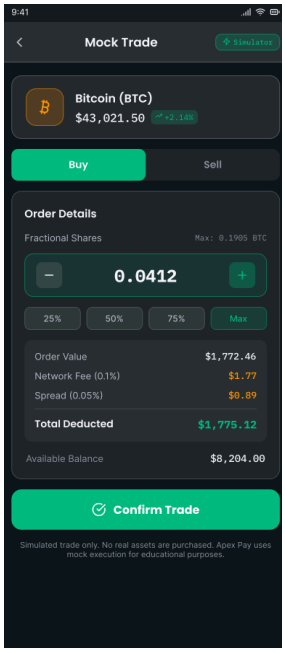
figma design :

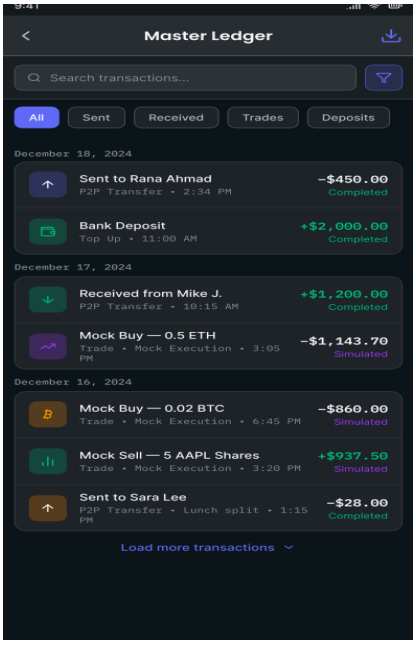


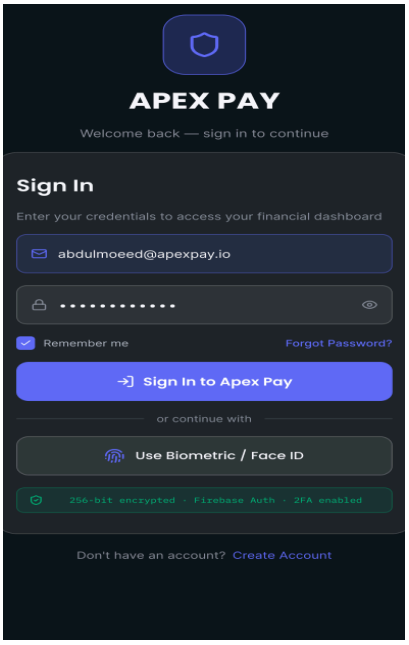

9:41
Apex AI Analyst
Online
Quick Questions
BTC NPV analysis
Diversify portfolio
DCF valuation
Should I invest more in BTC given current market volatility? What does the NPV analysis say?
9:38 AM
Based on DCF analysis, BTC shows a risk-adjusted NPV of +$4,200 over 12 months at your current position. Here's my assessment:
Risk Level MODERATE-HIGH
12M NPV Estimate +$4,200
Recommendation HOLD / ADD 10%
9:39 AM
Scan-to-Invest
Point camera at a brand logo to instantly map to its stock
Ask about risk, NPV, portfolio...
Home
Invest
Wallet
AI
Profile

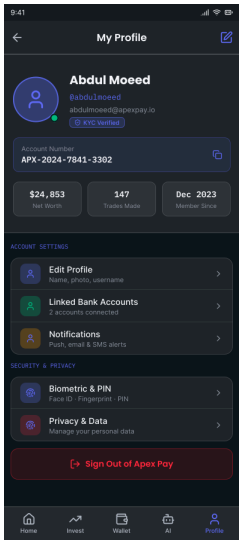

9:41
APEX PAY
Your financial ecosystem
Profile Photo
Tap to upload a photo
Choose
Email address
Choose a username
Confirm password
Password Strength: Strong
I agree to the Terms of Service & Privacy Policy
Create Secure Account
Already have an account? Sign In


9:41
APEX PAY
Welcome back, Abdul Moeed
Biometric Authentication
Use Face ID or fingerprint to access your account instantly
Use Face ID
Fingerprint
or use PIN
1
2
3
4
5
6
7
8
9


9:41
Mock Trade
Simulator
Bitcoin (BTC)
$43,021.50
+2.14%
Buy
Sell
Order Details
Fractional Shares
Max: 0.1905 BTC
0.0412
25%
50%
75%
Max
Order Value $1,772.46
Network Fee (0.1%) $1.77
Spread (0.05%) $0.89
Total Deducted $1,775.12
Available Balance $8,204.00
Confirm Trade
Simulated trade only. No real assets are purchased. Apex Pay uses mock execution for educational purposes.


Master Ledger
Search transactions...
All
Sent
Received
Trades
Deposits
December 18, 2024
Sent to Rana Ahmad
P2P Transfer • 2:34 PM
−$450.00
Completed
Bank Deposit
Top Up • 11:00 AM
+$2,000.00
Completed
December 17, 2024
Received from Mike J.
P2P Transfer • 10:15 AM
+$1,200.00
Completed
Mock Buy — 0.5 ETH
Trade • Mock Execution • 3:05 PM
−$1,143.70
Simulated
December 16, 2024
Mock Buy — 0.02 BTC
Trade • Mock Execution • 6:45 PM
−$860.00
Simulated
Mock Sell — 5 AAPL Shares
Trade • Mock Execution • 3:20 PM
+$937.50
Simulated
Sent to Sara Lee
P2P Transfer • Lunch split • 1:15 PM
−$28.00
Completed
Load more transactions


APEX PAY
Welcome back — sign in to continue
Sign In
Enter your credentials to access your financial dashboard
abdulmoeed@apexpay.io
Remember me
Forgot Password?
Sign In to Apex Pay
or continue with
Use Biometric / Face ID
256-bit encrypted · Firebase Auth · 2FA enabled
Don't have an account? Create Account


9:41
My Profile
Abdul Moeed
@abdulmoeed
abdulmoeed@apexpay.io
KYC Verified
Account Number
APX-2024-7841-3302
$24,853
Net Worth
147
Trades Made
Dec 2023
Member Since
ACCOUNT SETTINGS
Edit Profile
Name, photo, username
Linked Bank Accounts
2 accounts connected
Notifications
Push, email & SMS alerts
SECURITY & PRIVACY
Biometric & PIN
Face ID · Fingerprint · PIN
Privacy & Data
Manage your personal data
Sign Out of Apex Pay
Home
Invest
Wallet
AI
Profile